PROJECT INITIALIZATION AND PLANNING PHASE

Project Proposal (Proposed Solution) Template

Date	28 October 2026	Maximum Marks	3 Marks
Team ID	EN25CS3011166	Project Title	Health360: Cardiovascular Risk Analytics

Project Proposal Overview: This project proposal outlines a Business Intelligence solution to address cardiovascular risk assessment. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the methodology, key features, and resource requirements (hardware, software, data).

1. PROJECT OVERVIEW & PROBLEM STATEMENT

PROJECT OVERVIEW	
Objective	To develop an interactive, multi-dimensional Business Intelligence dashboard (Health360) that analyzes cardiovascular and stroke risk factors across patient cohorts, providing clinicians with actionable diagnostic risk heatmaps and risk correlations.
Scope	Integrates physiological dataset metrics (Age 19–79, Cholesterol 150–280 mg/dL, Blood Pressure 85–175 mmHg, BMI, Glucose) alongside lifestyle indicators (Smoking, Alcohol, Exercise Level) and clinical symptoms (Fatigue, Chest Pain, Dizziness) into unified Tableau/Streamlit visualizations.
PROBLEM STATEMENT	
Description	Cardiovascular diseases and stroke remain leading causes of mortality, yet traditional diagnostic workflows evaluate risk factors in isolation. Patients and clinicians lack unified tools to visualize how combined variables—such as elevated BMI with high Blood Pressure or Fatigue with Dizziness—escalate overall cardiac risk.
Impact	Delayed identification of high-risk cohorts leads to late clinical intervention, increased emergency hospitalizations, and preventable stroke occurrences. Providing automated risk heatmaps and multi-variable analytics enables early preventative care and personalized lifestyle modifications.
PROPOSED SOLUTION	
Approach	Implement an end-to-end analytical pipeline: Ingest clinical data into Snowflake/MySQL, perform calculated metric binning (Tableau Prep/Python), compute `AVG(Heart Disease)` and Health Risk Scores, and publish interactive visual dashboards accessible to healthcare providers.
Key Features	• **Age-Based Cardiac Risk Trends:** Visual line graphs for ages 19–79. • **Patient Risk Heatmap:** Binned Blood Pressure vs. BMI category matrix. • **Lifestyle Impact Bar Charts:** Alcohol & Smoking risk comparison and Exercise Level vs Risk. • **Symptom Correlation Panels:** Fatigue v/s Stroke Risk and Chest Pain v/s Dizziness. • **Interactive Parameter Filtering:** Dynamic filtering by Glucose, BMI, and Cholesterol ranges.

2. RESOURCE REQUIREMENTS

Resource Type	Description	Specification / Allocation
Hardware Resources		
Computing Resources	Standard development workstation / cloud instance for ETL execution and dashboard rendering.	**Intel Core i7 / 8 Cores CPU**
Memory	RAM for handling memory-intensive dataset transformations and Tableau extract generation.	**16 GB DDR4 RAM**
Storage	Disk space for storing raw CSV files, database schemas, tableau extracts (.hyper), and logs.	**512 GB NVMe SSD**
Software & Frameworks		
Frameworks	Web framework for hosting custom web interface dashboards.	**Streamlit / Flask (Python 3.11)**
Libraries	Data manipulation, numerical aggregation, and statistical plotting.	**Pandas, NumPy, Seaborn, Matplotlib**
Development Environment	BI visualization tool, SQL workspace, and code editor.	**Tableau Public / Desktop, VS Code, Git**